

Barycentric Maximal Rényi Divergence Is Strictly Smaller in Every Finite Dimension

Run `fa_banach_001`, agent `agent_lane_11`

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Source Conjecture and Result

Mosonyi, Bunth, and Vrana prove that the barycentric maximal Rényi divergence is bounded above by the maximal Rényi divergence. They conjecture that the inequality is strict for every noncommuting invertible pair, and prove this only when the underlying Hilbert space is two-dimensional [1, Sec. 6.4].

Source target (paraphrased). For $0 < \alpha < 1$ and noncommuting $\rho, \sigma > 0$, is $D_\alpha^{\text{bary, max}}(\rho \parallel \sigma) < D_\alpha^{\text{max}}(\rho \parallel \sigma)$ in every finite dimension?

We prove the conjecture. The source's chosen directional derivative is strictly negative in every dimension; the missing fact is that its scalar coefficient kernel is conditionally negative definite. Status: *candidate full solution, likely valid*, subject to human review.

Theorem 1. *Let ρ, σ be positive definite operators on a finite-dimensional complex Hilbert space and suppose $\rho\sigma \neq \sigma\rho$. Then, for every $\alpha \in (0, 1)$,*

$$D_\alpha^{\text{bary, max}}(\rho \parallel \sigma) < D_\alpha^{\text{max}}(\rho \parallel \sigma). \quad (1)$$

The Source Derivative

Put $d = \dim \mathcal{H}$ and choose an orthonormal eigenbasis for

$$R := \sigma^{-1/2} \rho \sigma^{-1/2} = \text{diag}(\lambda_1, \dots, \lambda_d).$$

Repeated eigenvalues are allowed, and let $S = (S_{ij})$ be the matrix of σ in this basis. The source evaluates the derivative of its barycentric variational objective along the segment from the normalized geometric-mean candidate to the maximally mixed state. Its formula is

$$\partial_\pi = -1 + \frac{1}{d} \sum_{i,j=1}^d S_{ij} (S^{-1})_{ji} \Lambda_{\alpha, i, j}, \quad (2)$$

where $\Lambda_{\alpha, i, i} = 1$ and, for $\lambda_i \neq \lambda_j$,

$$\Lambda_{\alpha, i, j} = \alpha(1 - \alpha)(\log \lambda_i - \log \lambda_j) \frac{\lambda_i - \lambda_j}{(\lambda_i^\alpha - \lambda_j^\alpha)(\lambda_i^{1-\alpha} - \lambda_j^{1-\alpha})}. \quad (3)$$

The source's variational reduction shows that $\partial_\pi < 0$ implies (1). It proves this sign from (2) only for $d = 2$.

Conditional Negative Definiteness

Recall that an even function $\psi : \mathbb{R} \rightarrow \mathbb{R}$ with $\psi(0) = 0$ is conditionally negative definite if, for all finite (x_i) and complex (c_i) with $\sum_i c_i = 0$,

$$\sum_{i,j} \bar{c}_i c_j \psi(x_i - x_j) \leq 0.$$

Lemma 1. *For every $\alpha \in (0, 1)$, the continuous even function*

$$\psi_\alpha(x) := \frac{\alpha(1-\alpha)x \sinh(x/2)}{2 \sinh(\alpha x/2) \sinh((1-\alpha)x/2)} - 1, \quad \psi_\alpha(0) := 0, \quad (4)$$

is conditionally negative definite, and $\psi_\alpha(x) > 0$ for $x \neq 0$.

Proof. Let $\beta = 1 - \alpha$ and $h(x) = (x/2) \coth(x/2)$, with $h(0) = 1$. The addition formula for $\sinh$ gives

$$\psi_\alpha(x) = \beta(h(\alpha x) - 1) + \alpha(h(\beta x) - 1). \quad (5)$$

The Mittag-Leffler expansion of $\coth$ yields the locally uniformly convergent series

$$h(x) - 1 = 2 \sum_{n=1}^{\infty} \frac{x^2}{x^2 + 4\pi^2 n^2}. \quad (6)$$

For $a > 0$,

$$\frac{x^2}{x^2 + a^2} = a \int_0^\infty e^{-at} (1 - \cos(tx)) dt \quad (7)$$

is conditionally negative definite: when $\sum_i c_i = 0$, the corresponding quadratic form is

$$-a \int_0^\infty e^{-at} \left| \sum_i c_i e^{itx_i} \right|^2 dt \leq 0.$$

Positive sums, locally uniform limits, rescalings, and positive linear combinations preserve conditional negative definiteness. Equations (5)–(7) prove the first assertion. Every summand in (6) is positive away from zero, so the second assertion follows as well. $\square$

A Double-Commutator Sign Lemma

Lemma 2. *Let $S > 0$ be a $d \times d$ matrix and let $\Psi = (\Psi_{ij})$ have the form $\Psi_{ij} = \|v_i - v_j\|^2$ for real vectors $v_1, \dots, v_d$. Then*

$$\mathrm{Tr} [S^{-1}(S \circ \Psi)] \leq 0, \quad (8)$$

where $\circ$ is the Schur product. Equality holds exactly when $S_{ij} = 0$ whenever $v_i \neq v_j$.

Proof. Writing $v_i = (v_{ik})_k$, set $V_k = \mathrm{diag}(v_{1k}, \dots, v_{dk})$. Entrywise,

$$S \circ \Psi = \sum_k [V_k, [V_k, S]]. \quad (9)$$

Diagonalize S as $Su_a = s_a u_a$, $s_a > 0$. For any self-adjoint V , a direct expansion gives the quantitative identity

$$\mathrm{Tr} S^{-1} [V, [V, S]] = -2 \sum_{a < b} \frac{(s_a - s_b)^2}{s_a s_b} |\langle u_a, V u_b \rangle|^2 \leq 0. \quad (10)$$

Equality in (10) is equivalent to $[S, V] = 0$. Summing over k proves (8); simultaneous equality says $(v_{ik} - v_{jk})S_{ij} = 0$ for every i, j, k , which is the stated condition. $\square$

Proof of Theorem 1

Proof. Set $x_i = \log \lambda_i$. Rewriting (3) in hyperbolic form shows

$$\Lambda_{\alpha,i,j} - 1 = \psi_\alpha(x_i - x_j). \quad (11)$$

By Lemma 1, the finite matrix $\Psi_{ij} := \psi_\alpha(x_i - x_j)$ is a squared-distance matrix: explicitly, with $J = I - d^{-1}\mathbf{1}\mathbf{1}^*$, $G = -\frac{1}{2}J\Psi J \geq 0$; factoring G as a Gram matrix produces real vectors v_i satisfying $\Psi_{ij} = \|v_i - v_j\|^2$.

Since

$$\sum_{i,j} S_{ij}(S^{-1})_{ji} = \text{Tr}(SS^{-1}) = d,$$

equations (2) and (11) give

$$d \partial_\pi = \sum_{i,j} S_{ij}(S^{-1})_{ji} \Psi_{ij} = \text{Tr}[S^{-1}(S \circ \Psi)] \leq 0 \quad (12)$$

by Lemma 2.

It remains to rule out equality. Lemma 1 says $v_i = v_j$ exactly when $x_i = x_j$, equivalently $\lambda_i = \lambda_j$. Equality in (12) would therefore force $S_{ij} = 0$ whenever $\lambda_i \neq \lambda_j$. Thus S would commute with $\text{diag}(\lambda_i)$, i.e. $[\sigma, R] = 0$. Because $R = \sigma^{-1/2} \rho \sigma^{-1/2}$, this is equivalent to $[\rho, \sigma] = 0$, contrary to the hypothesis. Hence $\partial_\pi < 0$.

The normalized geometric-mean candidate and the maximally mixed state are both positive definite states. Therefore a sufficiently short positive step along the source's segment strictly lowers its variational objective. The source reduction following (2) now yields (1). $\square$

Novelty Check and Verification

The cheap run indexes contained no answer. Bounded searches through 17 August 2026 used the exact conjecture, title, arXiv id, citations, and the terms *barycentric maximal Rényi*, *strict inequality*, and *conditionally negative definite*. They found arXiv v5, the 2024 publication, and later applications, but no extension of the two-dimensional theorem. This is not a priority claim.

The companion script checks the hyperbolic decomposition, a 20,000-term partial-fraction approximation, finite Schoenberg embeddings, the double-commutator identity, commuting equality cases, and 5,280 random strict derivative instances in dimensions 2 through 12. These checks are auxiliary; the proof above is exact.

Human Review Recommendation

Verify (5) and the normalization in (2). Then check the transpose convention in (12) and the equality implication $[S, V_k] = 0$ for every k . Those are the only nonstandard steps.

References

- [1] M. Mosonyi, G. Bunth, and P. Vrana, *Geometric relative entropies and barycentric Rényi divergences*, Linear Algebra Appl. **699** (2024), 159–276; arXiv:2207.14282v5.